RH124 — Chapter 15

find & grep Examples

Using /etc/passwd as the target file

Part 1 — Understanding /etc/passwd

Before using grep and find, students must understand the file structure:

```
# /etc/passwd format: 7 fields separated by :
ali:x:1001:1001:Ali Hassan:/home/ali:/bin/bash
 ^username  ^pwd  ^UID   ^GID   ^comment      ^home          ^shell
```

> **Note:** Every user account has one line in /etc/passwd. The file is world-readable — anyone can view it, but passwords are stored in /etc/shadow (root only).

Viewing /etc/passwd

```
$ cat /etc/passwd
  # view entire file

$ cat /etc/passwd | head -5
  # show first 5 lines only

$ tail -3 /etc/passwd
  # show last 3 lines (recently added users)

$ wc -l /etc/passwd
  # how many users/lines in the file?
  56
```

Part 2 — grep Examples on /etc/passwd

grep searches inside file content — use it to find specific users, shells, or patterns.

Basic grep — Find a User

```
$ grep ali /etc/passwd
  # find lines containing 'ali'
  ali:x:1001:1001:Ali Hassan:/home/ali:/bin/bash

$ grep root /etc/passwd
```

```
  root:x:0:0:root:/root:/bin/bash
  operator:x:11:0:operator:/root:/sbin/nologin
  # finds ALL lines containing 'root'
```

grep with Flags

-i Case-insensitive search

```
$ grep -i ALI /etc/passwd
  # finds: ali, Ali, ALI — ignores case
  ali:x:1001:1001:Ali Hassan:/home/ali:/bin/bash
```

-n Show line numbers

```
$ grep -n bash /etc/passwd
  # shows line number where bash appears
  1:root:x:0:0:root:/root:/bin/bash
  42:ali:x:1001:1001:Ali:/home/ali:/bin/bash
```

-v Invert match — show lines that do NOT match

```
$ grep -v nologin /etc/passwd
  # show users who CAN log in (no nologin shell)

$ grep -v "^#" /etc/passwd
  # skip comment lines (lines starting with #)
```

-c Count matching lines

```
$ grep -c nologin /etc/passwd
  # how many system accounts (nologin)?
  38

$ grep -c bash /etc/passwd
  # how many users have bash shell?
  4
```

-w Match whole word only

```
$ grep -w ali /etc/passwd
  # matches 'ali' but NOT 'alicia' or 'somali'
```

^ Match at start of line (regex)

```
$ grep "^root" /etc/passwd
  # only lines that START with 'root'
  root:x:0:0:root:/root:/bin/bash
```

```
$ grep "^ali" /etc/passwd
  ali:x:1001:1001:Ali Hassan:/home/ali:/bin/bash
```

$ and special shell patterns

```
$ grep "bash$" /etc/passwd
  # lines that END with 'bash' (bash shell users)

$ grep "nologin$" /etc/passwd
  # all service accounts (end with nologin)
```

grep + cut — Extract Specific Fields

Combine grep with cut to extract only the fields you need from /etc/passwd:

```
$ grep bash /etc/passwd | cut -d: -f1
  # get usernames of all bash users
  # -d: = delimiter is ':'   -f1 = field 1 (username)
  root
  ali
  sara

$ grep ali /etc/passwd | cut -d: -f1,3,7
  # get username:UID:shell for ali
  ali:1001:/bin/bash

$ cut -d: -f1 /etc/passwd
  # list ALL usernames on the system
```

grep Quick Reference

Command / Flag	Description
`grep pattern file`	Search for pattern in file
`grep -i pattern file`	Case-insensitive search
`grep -n pattern file`	Show line numbers
`grep -v pattern file`	Show lines NOT matching
`grep -c pattern file`	Count matching lines
`grep -w pattern file`	Match whole word only
`grep -r pattern dir/`	Search recursively in directory
`grep -l pattern dir/`	List files that contain match
`grep '^text' file`	Lines starting with text
`grep 'text$' file`	Lines ending with text

Command / Flag	Description
`grep -E 'a|b' file`	Extended regex — match a OR b

Part 3 — find Examples

find searches for FILES by name, owner, size, type, permissions — not file content.

> **Key difference:** grep searches INSIDE files (content). find searches FOR files (by attributes like name, owner, size, date).

Find Files by Name

```
$ find /etc -name "passwd"
  # find file named exactly 'passwd' in /etc
  /etc/passwd

$ find /etc -name "pass*"
  # wildcard — any file starting with 'pass'
  /etc/passwd
  /etc/passwd-

$ find /etc -name "*.conf"
  # find all .conf files in /etc

$ find /etc -iname "passwd"
  # -iname = case-insensitive (Passwd, PASSWD, passwd)
```

Find Files by Owner

```
$ find /home -user ali
  # all files owned by ali in /home

$ find / -user ali 2>/dev/null
  # find all of ali's files on entire system
  # 2>/dev/null hides 'permission denied' errors

$ find /home -group developers
  # files owned by group developers

$ find / -nouser  2>/dev/null
  # orphan files — no owner (user was deleted)
```

Find Files by Type

```
$ find /etc -type f
  # f = regular files only

$ find /etc -type d
  # d = directories only

$ find /etc -type l
  # l = symbolic links only

$ find /etc -type f -name "*.conf"
  # combine: regular .conf files only
```

Find Files by Size

```
$ find /var -size +100M
  # files LARGER than 100 MB

$ find /tmp -size -1k
  # files SMALLER than 1 KB

$ find / -size +1G 2>/dev/null
  # files larger than 1 GB (disk cleanup!)

  # Size units: c=bytes  k=KB  M=MB  G=GB
  # + = greater than    - = less than    (none) = exact
```

Find Files by Time

```
$ find /etc -mtime -7
  # modified in the LAST 7 days

$ find /var/log -mtime +30
  # modified MORE than 30 days ago (old logs)

$ find /home -newer /etc/passwd
  # files newer than /etc/passwd

  # -mtime = modify time    -atime = access time
  # -ctime = change time    -newer = newer than file
```

Find Files by Permissions

```
$ find / -perm /4000 2>/dev/null
  # find all SUID files (security audit!)
```

```
$ find / -perm /2000 2>/dev/null
  # find all SGID files

$ find /home -perm 777
  # find world-writable files (security risk!)
```

find with Actions — Execute Commands

```
$ find /tmp -name "*.log" -delete
  # find and DELETE all .log files in /tmp

$ find /home/ali -type f -exec ls -lh {} \;
  # run 'ls -lh' on every file found
  # {} = placeholder for found file    \; = end of command

$ find /home -user ali -exec chown sara {} \;
  # find ali's files and change owner to sara

$ find /var/log -mtime +90 -exec rm {} \;
  # delete log files older than 90 days
```

Combining find and grep Together

Use find to locate files, then pipe to grep to search inside them:

```
$ find /etc -name "*.conf" -exec grep -l "sshd" {} \;
  # find .conf files that mention 'sshd'

$ grep -r  "PermitRootLogin" /etc/ssh/
  # -r = recursive grep (replaces find+grep)

$ find / -name "passwd" 2>/dev/null
  /etc/passwd
  /etc/pam.d/passwd
```

find Quick Reference

Command / Flag	Description
`find /path -name 'f'`	Find by exact filename
`find /path -iname 'f'`	Find by name (case-insensitive)
`find /path -user ali`	Find files owned by user ali

Command / Flag	Description
`find /path -group grp`	Find files owned by group
`find /path -nouser`	Find orphan files (no owner)
`find /path -type f`	Regular files only
`find /path -type d`	Directories only
`find /path -type l`	Symbolic links only
`find /path -size +100M`	Files larger than 100 MB
`find /path -mtime -7`	Modified in last 7 days
`find /path -mtime +30`	Modified more than 30 days ago
`find /path -perm /4000`	Files with SUID bit set
`find /path -perm 777`	Files with exact permissions
`find /path -exec cmd {} \;`	Run command on each result
`find /path -delete`	Delete matching files

Part 4 — Exam Tips

Tip 1: grep searches INSIDE file content. find searches FOR files by attributes. Never mix them up!

Tip 2: Always add 2>/dev/null to find commands on / — hides 'permission denied' errors from restricted directories.

Tip 3: grep -v nologin /etc/passwd shows users who CAN log in interactively. Very useful for security audits.

Tip 4: find / -nouser 2>/dev/null finds orphaned files after a user is deleted. Always clean these up!

Tip 5: Use grep '^username' (with ^) to find exact username matches — without ^ it might match 'username2' or 'myusername'.

Warning: find -exec rm {} \; is PERMANENT — always run without -exec first to preview what will be deleted!